Randall Krouth

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Randall Krouth's Portfolio | Credly Badges | LinkedIn

Professional Summary

Full-stack developer with strong experience in .NET, cross-platform application development, web development, cloud services, machine learning, cybersecurity, GPU accelerated computing, and secure software engineering. Skilled in scalable architecture, RESTful API Development, CI/CD pipelines, ML model design, and programming high-performance systems. Strong foundation in parallel computing, memory optimizations, DevSecOps, and penetration testing.

Manufacturing Technician with strong expertise in automated MIG welding, CNC machine operation, and preventative maintenance protocols. Adept at managing quick line changeovers, monitoring cell efficiency, and conducting hands-on quality checks to ensure structural integrity and zero-defect manufacturing.

Education

SUNY Brockport – Brockport, NY

B.S. Computer Science (Major) | Cybersecurity (Minor)

Technical Skills

• **Languages:** .NET, C#, C++, C, Java, Python, JavaScript, TypeScript, SQL, Swift, HTML/CSS, XAML, YAML, XML, JSON

• **Frameworks/Tools:** NGINX, Syncfusion, Qt, CUDA, OpenCL, Apache Arrow, Keycloak, Auth0, Jenkins, Docker, Kubernetes, FastAPI, Django, Flask, Express.js, ML.NET, PyTorch, TensorFlow, Kestrel, React, Angular, Node.js, JUCE, Unity, Unreal Engine, NoSQL, MongoDB, Kali Linux, Kafka, gRPC, Websockets, NATS, QUIC, Tableau, Power BI, Plotly, OpenShift, JWT's, LDAP, PostgreSQL

• **Cloud Service Providers**: Microsoft Azure, Amazon Web Services, Google Cloud Platform

• **Strengths:** Secure coding practices, parallel processing, asynchronous programming, memory and cache optimizations, prompt engineering, user interface development, and hypervisor virtual machine usage including VMware, Virtual Box, and various cloud instances.

Professional Experience

Independent Software Development Projects, *2021-Present*

- Designed & developed full-stack applications including multi-tenant architectures through ASP.NET Core Web API, Docker, and cloud technologies.
- Built projects focused on machine learning, web applications, API development, and cybersecurity.
- Developed REST APIs, front-end systems, back-end systems, and CI/CD pipelines.
- Achieved 99.8% security scanner capability across publicly exposed benchmark frameworks by improving API Tester headless covering OWASP, XBOW, and various benchmark frameworks.
- Designed scalable architecture across ML, API, and game systems.

Technical Projects

Automated Cloud Coding Environment, *2026-Present*

- Developing a cloud multi-tenant SaaS cloud coding and agentic autonomous development environment.
- Application analyzes system files, automates processes, performs security testing, and automates CI/CD pipelines with agentic oversight.

Trading Platform, *2024-Present*

- Developed front-end in .NET Maui + Blazor, created backend utilizing ASP .NET Core, and Python Uvicorn servers to obtain a scalable multi-tenant architecture.
- Created custom database integration using API development, DuckDB adapter, and SQLite registry for unparalleled Apache Arrow speed & performance.
- Integrated machine learning models for predictive analytics and data processing.

Certifications

• IBM: Full Stack Software Development, Back-End Development, AI Engineering, IT Fundamentals for Cybersecurity, DevOps & Software Engineering, Security Analyst

• Google: UX Design, Advanced Data Analytics, Cloud Fundamentals: Core Infrastructure

• Security & Cloud: LearnKartS Certified Ethical Hacking, TryHackMe DevSecOps, TryHackMe Security Engineer, LearnKartS Certified Solutions Architect, Whizlabs Azure Identity Governance and Monitoring Solutions, UC Davis Secure Coding Practices

Training & Additional Competencies

• Agile development & scrum

• Database Administration: PostgreSQL, NoSQL, MySQL, Db2, MongoDB

• Predictive analytics & hyperparameter tuning, application optimizations, performance/load testing, diagnostics, and security hardening.

Professional Experience

Precise Tool & Manufacturing, Inc., *2025-2026*

Executed high-precision CNC machining operations, maintaining tight tolerances and strict adherence to engineering blueprints and manufacturer specifications. Conducted rigorous quality control inspections using precision measuring instruments (calipers, micrometers, height gauges) to verify part dimensions. Performed post-machining deburring and surface finishing to remove imperfections, ensuring flawless edge quality and structural integrity.

IDEX Health & Science, *2024-2025*

Executed preventative maintenance protocols on optical coaters to minimize downtime, ensure peak operational efficiency, and extend equipment life. Performed rapid machine changeovers and set-ups, configuring tooling and fixturing for precision drilling and machining runs. Inspected produced components against strict manufacturer standards, catching micro-defects early to uphold high-quality standards in a regulated manufacturing environment.

McAlpin Industries, Inc., *2023-2024*

Operated automated robotic welding cells to execute continuous, high-efficiency MIG welding on production components. Performed manual touch-up MIG welds and quality checks to ensure full weld penetration, clean bead profiles, and structural compliance. Inspected completed assemblies against manufacturer specifications, identifying seam misalignments and preventing defective parts from moving downstream.